

Test Summary Report

Project: AI Chatbot Management System for Facebook Messenger **Author:** Nguyễn Ngọc Minh — 22110154 **Test Period:** May – June 2026 **Test Method:** Structured Manual Testing **Environment:** Windows 11, Android Emulator Pixel 3 API 34, Python 3.13

Executive Summary

The system was tested across 48 structured test cases covering 7 functional groups. All 48 test cases passed (100% pass rate). Additional qualitative evaluation was performed on AI response quality in Vietnamese. Five known limitations were identified and documented for future improvement.

Metric	Value
Total Test Cases	48
Passed	48
Failed	0
Pass Rate	100%
Groups Covered	7
Qualitative Evaluations	1 (AI Quality Assessment)
Known Limitations	5

1. Test Scope & Strategy

1.1 What Was Tested

- **Authentication flow** — login, token storage, session expiry, logout
- **Bot management (CRUD)** — create, read, toggle, delete, search
- **Automated message processing** — webhook → AI → reply (end-to-end)
- **Chat history display** — bubble UI, timestamps, pagination, empty/error states
- **Statistics dashboard** — metrics accuracy, chart rendering, zero-data handling
- **API security** — Bearer token, webhook signature, encrypted storage
- **UI consistency** — loading states, confirmation dialogs, screen sizes

1.2 What Was NOT Tested

- **Automated unit tests** — all testing was manual; no JUnit or pytest suites
- **Load/stress testing** — concurrent webhook volume was not benchmarked
- **Performance benchmarks** — AI response time was observed but not formally measured
- **Cross-device testing** — tested only on Pixel 3 emulator (API 34)

- **Facebook API edge cases** — sticker-only messages, group chats, page mentions

1.3 Test Tools

Tool	Purpose
Swagger UI	Manual API endpoint testing
Android Emulator (Pixel 3 API 34)	Application UI and functional testing
Facebook Messenger (Mobile)	End-to-end chatbot conversation testing
ngrok	Public URL for webhook connectivity
Logcat	Android runtime error monitoring
OkHttp Logging Interceptor	HTTP request and response debugging
DB Browser for SQLite	Direct database verification

2. Test Results by Group

2.1 Authentication Module (TC-01 to TC-06)

Objective: Verify login, token management, and session handling.

Test Case ID	Test Case Description	Status
TC-01	Login with valid credentials	✓ Pass
TC-02	Login with invalid password	✓ Pass
TC-03	Login with empty fields	✓ Pass
TC-04	Token storage after login	✓ Pass
TC-05	Invalid token handling	✓ Pass
TC-06	Logout functionality	✓ Pass
Result	Passed / Total	6 / 6

2.2 Bot Management Module (TC-07 to TC-15)

Objective: Verify all CRUD operations and search functionality.

Test Case ID	Test Case Description	Status
TC-07	Create bot with valid data	✓ Pass
TC-08	Create bot with missing fields	✓ Pass
TC-09	Create bot with duplicate Page ID	✓ Pass
TC-10	Enable disabled bot	✓ Pass
TC-11	Disable enabled bot	✓ Pass
TC-12	Delete bot	✓ Pass
TC-13	Search by bot name	✓ Pass

TC-14	Search by Page ID	✓ Pass
TC-15	Swipe to refresh	✓ Pass
Result	Passed / Total	9 / 9

2.3 Automated Message Processing (TC-16 to TC-23)

Objective: Verify the complete webhook → AI → reply flow.

Test Case ID	Test Case Description	Status
TC-16	Active bot replies to message	✓ Pass
TC-17	Disabled bot ignores messages	✓ Pass
TC-18	Typing indicator display	✓ Pass
TC-19	Multi-turn conversation context	✓ Pass
TC-20	Message persistence in database	✓ Pass
TC-21	Ollama fallback handling	✓ Pass
TC-22	Echo message filtering	✓ Pass
TC-23	Attachment and sticker handling	✓ Pass
Result	Passed / Total	8 / 8

Notes on TC-19: Multi-turn context works at a technical level (database retrieves correct messages per sender_id). However, the Phi-3 model's ability to use that context is limited — see Section 4 for AI quality evaluation.

Notes on TC-21: Fallback messages rotate between 3 templates to avoid repetition. Verified by stopping Ollama and sending 3 consecutive messages — each received a different fallback text.

2.4 Chat History Module (TC-24 to TC-30)

Objective: Verify conversation display, formatting, and state handling.

Test Case ID	Test Case Description	Status
TC-24	Chronological display order	✓ Pass
TC-25	Bubble differentiation	✓ Pass
TC-26	Timestamp formatting	✓ Pass
TC-27	Empty state handling	✓ Pass
TC-28	Refresh conversation history	✓ Pass

TC-29	Error state handling	✓ Pass
TC-30	Pagination support	✓ Pass
Result	Passed / Total	7 / 7

2.5 Statistics Module (TC-31 to TC-35)

Objective: Verify dashboard metrics and chart rendering accuracy.

Test Case ID	Test Case Description	Status
TC-31	Statistics with real data	✓ Pass
TC-32	Seven-day chart rendering	✓ Pass
TC-33	Hourly distribution chart	✓ Pass
TC-34	Statistics with zero data	✓ Pass
TC-35	Peak-hour calculation	✓ Pass
Result	Passed / Total	5 / 5

Verification method: For TC-31, the displayed values were cross-referenced against direct SQL queries on the bots.db file using DB Browser for SQLite. All counts matched.

2.6 API Security Module (TC-36 to TC-41)

Objective: Verify authentication and authorization mechanisms.

Test Case ID	Test Case Description	Status
TC-36	API request without token	✓ Pass
TC-37	API request with invalid token	✓ Pass
TC-38	API request with valid token	✓ Pass
TC-39	Webhook without signature	✓ Pass
TC-40	Webhook with fake signature	✓ Pass
TC-41	Webhook with valid signature	✓ Pass
Result	Passed / Total	6 / 6

2.7 UI Consistency Module (TC-42 to TC-48)

Objective: Verify visual states, dialogs, and responsiveness.

Test Case ID	Test Case Description	Status
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TC-42	Loading ProgressBar display	✓ Pass
TC-43	Login button disabled during authentication	✓ Pass
TC-44	Empty bot list state	✓ Pass
TC-45	Delete confirmation dialog	✓ Pass
TC-46	Disable bot confirmation dialog	✓ Pass
TC-47	5-inch screen layout	✓ Pass
TC-48	6.5-inch screen layout	✓ Pass
Result	Passed / Total	7 / 7

3. Test Coverage Matrix

Requirement	Test Cases	Coverage
User Authentication	TC-01 – TC-06	✓ Full
Bot CRUD Operations	TC-07 – TC-15	✓ Full
Automated Message Reply	TC-16 – TC-23	✓ Full
Chat History Display	TC-24 – TC-30	✓ Full
Statistics Dashboard	TC-31 – TC-35	✓ Full
Webhook Signature Verification	TC-39 – TC-41	✓ Full
Bearer Token Authentication	TC-36 – TC-38	✓ Full
Encrypted Token Storage	TC-04	⚠ Partial
AI Response Quality (Vietnamese)	Qualitative Evaluation	⚠ Partial
Concurrent Webhook Handling	Not Tested	✗ None
Performance / Load Testing	Not Tested	✗ None
Cross-Device Compatibility	TC-47, TC-48 Only	⚠ Partial

4. AI Response Quality Evaluation

A qualitative evaluation of chatbot response quality was conducted using real Vietnamese-language conversations on Facebook Messenger with two bots: "BOT dịch vụ" (service advisor) and "Quản lý tài chính" (financial advisor).

4.1 Evaluation Criteria

Criteria	Description	Rating
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Response Language	Does the chatbot respond in Vietnamese?	🔴 Weak
System Prompt Adherence	Does the chatbot remain within its assigned role?	🟡 Medium
Response Relevance	Does the response address the user's question?	🔴 Weak
Context Retention	Does the chatbot remember previous messages?	🟡 Medium
Fallback Mechanism	Does the chatbot handle AI failures gracefully?	🟢 Good

4.2 Detailed Observations

Turn	User Input	Bot Response Summary	Assessment
1	"xin chào"	Replied in English and explained how to translate "hello" into Vietnamese	❌ Wrong language, irrelevant response
2	"cho tôi lấy vé xếp hàng y tế"	Continued discussing Vietnamese greetings	❌ Ignored user request
3	"tôi muốn hỏi về chứng khoán"	Responded in Vietnamese about stocks but with rambling content	⚠️ Partially relevant
4	"alo"	Continued previous stock discussion with incoherent content	⚠️ Context recognized but poorly utilized
5	(Ollama overloaded)	"Xin lỗi, mình đang bận xử lý nhiều yêu cầu"	✅ Fallback mechanism worked correctly

4.3 Root Cause Analysis

The quality issues stem from the **Phi-3 Mini 3.8B model itself**, not from the system architecture:

- Phi-3 is trained predominantly on English data — Vietnamese generation is limited
- Small model size (3.8B) reduces instruction-following capability compared to 7B+ models

- The system correctly retrieves context per sender_id from the database — the context mechanism works; the model's ability to utilize that context is the bottleneck

4.4 Recommended Improvements

Solution	Implementation Effort	Expected Impact
Switch to Llama 3.1 8B (Vietnamese-tuned)	Low — change one line in .env	High
Add explicit Vietnamese-only instructions in system prompt	Low — modify prompt text	Medium
Implement Retrieval-Augmented Generation (RAG) with business documents	High — requires architectural changes	Very High

5. Security Test Results

5.1 Webhook Signature Verification

Test Scenario	Method	Result
Missing Signature Header	POST /webhook without X-Hub-Signature-256	HTTP 403 
Invalid Signature	POST /webhook with fabricated signature	HTTP 403 
Valid Signature	POST /webhook with correct HMAC-SHA256 signature	HTTP 200 
Timing Attack Resistance	Uses hmac.compare_digest() constant-time comparison	Verified 

5.2 API Token Authentication

Test Scenario	Method	Result
Missing Authorization Header	GET /bots without Authorization header	HTTP 401 
Invalid Token	Bearer xyz123	HTTP 401 
Valid Token	Bearer correct-token	HTTP 200 
Expired Session Handling	HTTP 401 returned to Android application	Redirected to Login Screen 

5.3 Encrypted Storage Verification

Test Scenario	Method	Result
Token Stored Encrypted	Inspect <code>secure_auth_prefs.xml</code> using Device File Explorer	Encrypted ciphertext stored 
Token Removal on Logout	Execute <code>SecurePrefs.clear()</code> and inspect file	Token removed successfully 
Token Protection Verification	Read XML file directly on device	Only Base64 ciphertext visible, no plaintext token 